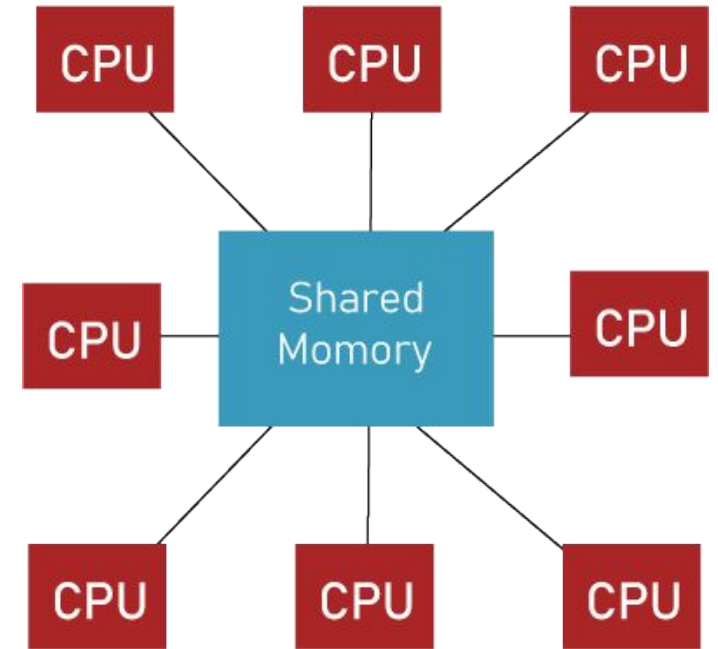


Multiprocessor and Multicomputers

A **Multiprocessor** is a computer system with two or more CPUs share full access to a common RAM.



- The main objective of using a multiprocessor is to boost the system's execution speed.
- There are two types of multiprocessors, one is called shared memory multiprocessor and another is distributed memory multiprocessor.
- **In shared memory multiprocessors**, all the CPUs shares the common memory.
- **In a distributed memory multiprocessor**, every CPU has its own private memory.

The interconnection among two or more processor and shared memory is done with three methods

1)Time shared common

bus 2)Multiport memories

3)Crossbar switch network

Time Shared Common Bus

in this method it contains a single shared bus through which all processor & memory unit can be communicated.

Consider CPU-1 is interacting with memory unit using common shared bus in that case all other processor must be idle as we have only one bus to communicate.

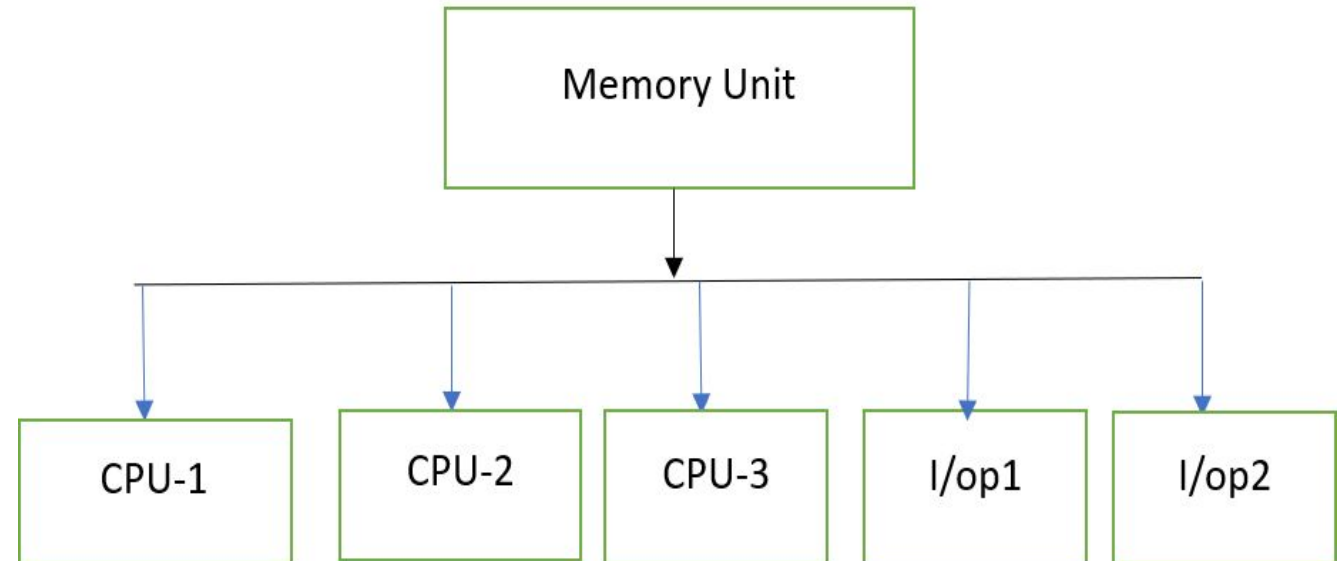
Advantage:

Simple to implement.

Due to single common bus cost to implement is very less.

Disadvantage:

Data transfer rate is slow.

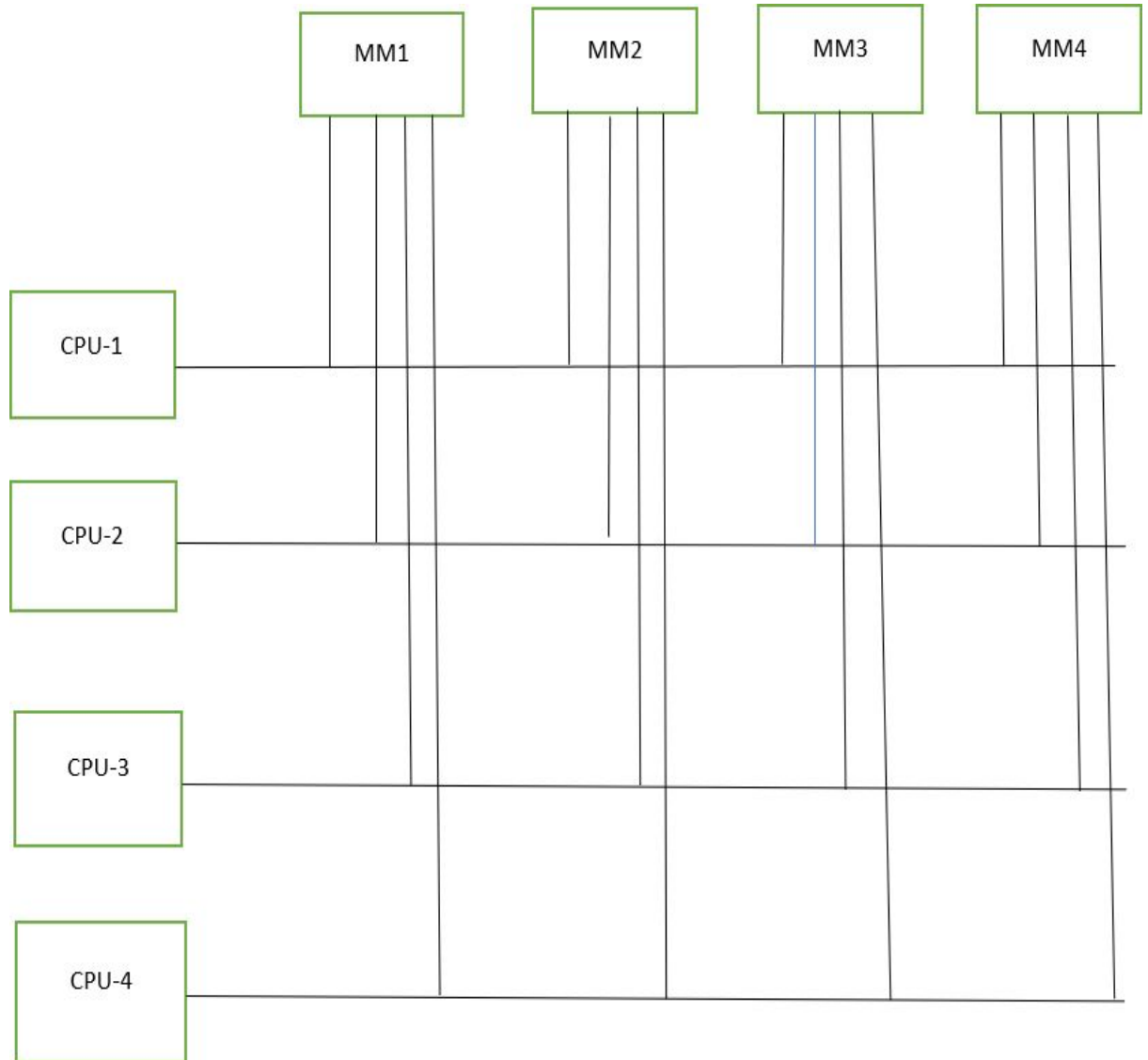


Multiport Memories

It contains separate bus for each processor to communicate with the memory module.

Suppose CPU-1 wants to interact with memory module 1 then port mm1 is enabled. Similarly CPU-4 wants to interact with memory module 4 then port mm4 is enabled. Hence all the process can be communicated parallelly.

If more than one CPU request for same time memory module, priority will be given in the order of CPU-1,CPU-2,CPU-3,CPU-4.



Crossbar Switch Network

A switch is installed between memory unit and CPU. Switch is responsible for whether to pass the request to a particular memory module or not based on the request made for.

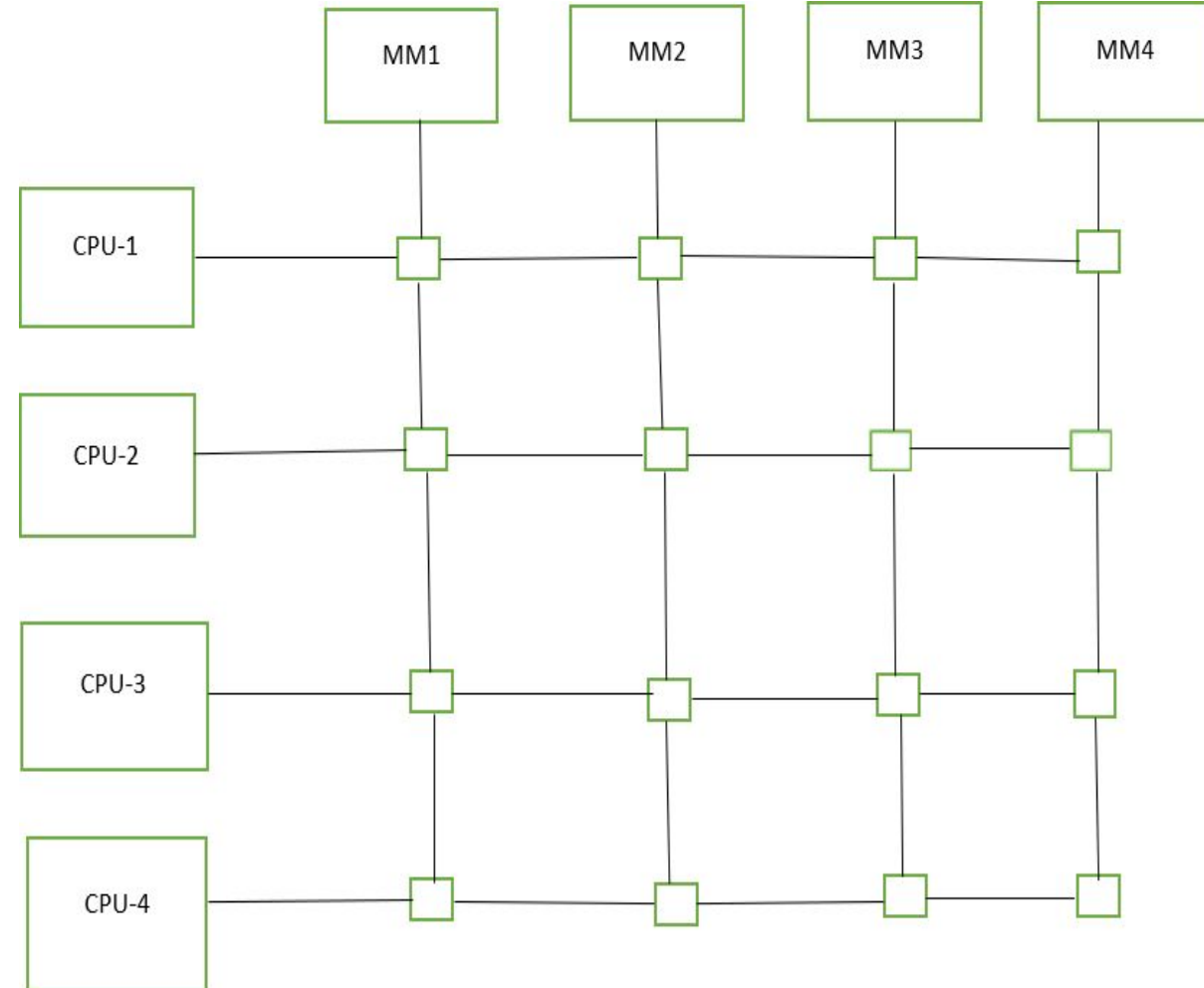
Advantage:

High data through rate.

Disadvantage:

Complex to implement as more switches involved.

Costlier to implement.



Applications of Multiprocessor –

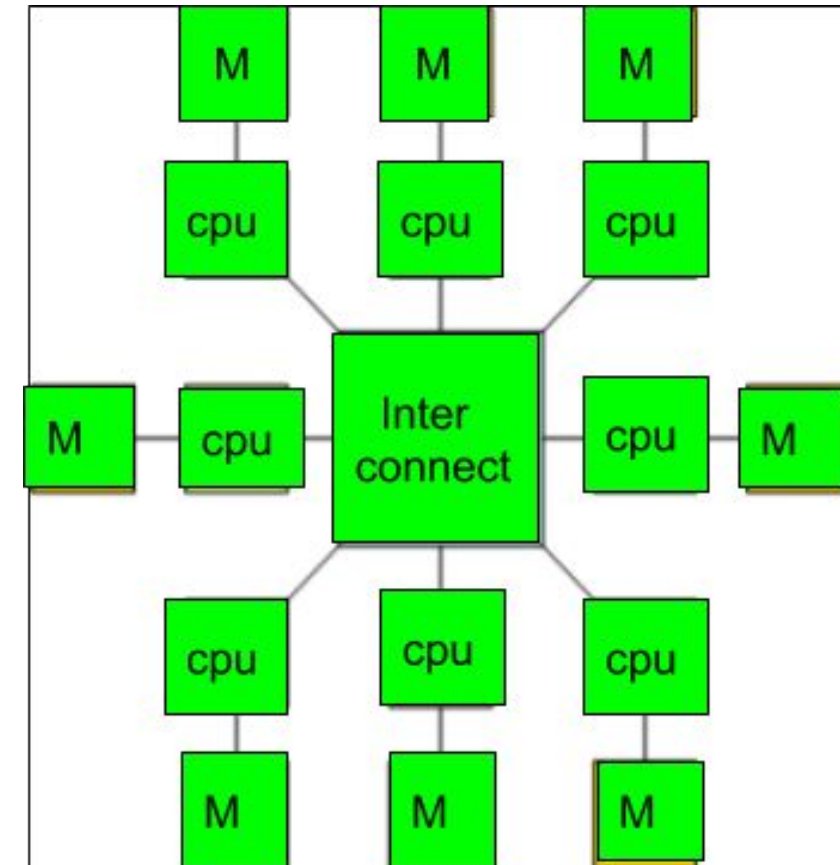
- As a uniprocessor, such as single instruction, single data stream (SISD).
- As a multiprocessor, such as single instruction, multiple data stream (SIMD), which is usually used for vector processing.
- Multiple series of instructions in a single perspective, such as multiple instruction, single data stream (MISD), which is used for describing hyper-threading or pipelined processors.
- Inside a single system for executing multiple, individual series of instructions in multiple perspectives, such as multiple instruction, multiple data stream (MIMD).

Advantages:

- Improved performance: Multiprocessor systems can execute tasks faster than single-processor systems, as the workload can be distributed across multiple processors.
- Better scalability: Multiprocessor systems can be scaled more easily than single-processor systems, as additional processors can be added to the system to handle increased workloads.
- Increased reliability: Multiprocessor systems can continue to operate even if one processor fails, as the remaining processors can continue to execute tasks.
- Reduced cost: Multiprocessor systems can be more cost-effective than building multiple single-processor systems to handle the same workload.
- Enhanced parallelism: Multiprocessor systems allow for greater parallelism, as different processors can execute different tasks simultaneously.

Multicompute

- A multicomputer system is a computer system with multiple processors that are connected together to solve a problem.
- Each processor has its own memory and it is accessible by that particular processor and those processors can communicate with each other via an interconnection network.
- **As the multicomputer is capable of messages passing between the processors, it is possible to divide the task between the processors to complete the task.** Hence, a multicomputer can be used for distributed computing.
- It is cost effective and easier to build a multicomputer than a multiprocessor.



Multiprocessor	Multicomputer
Multiprocessor consists of multiple processors within a single computer.	Multicomputer is an interlinked multiple autonomous computer.
Multiprocessor is a singly shared memory that is attached to the elements being processed.	The memory attached to the processing elements are distributed in multiples.
Multiprocessor is necessary for the processing elements to communicate with each other.	Multicomputer required message passing for elements being processed to communicate.
Multiprocessor is a dynamic network.	Multicomputer is a type of static network.
Example of multiprocessor is a sequent symmetry S-81.	Example of a multicomputer is a message passing multicomputer.

Three shared memory multiprocessor models are as follows –

- UMA Model
- UMA stands for Uniform memory access Model.
- NUMA model
- NUMA stands for non-uniform memory access.
- COMA model
- COMA stands for cache only memory access